STANDATA interpretation 23-BCI-023

Building

“Building Envelope” definition related to attic or roof spaces

Date Issued: January 2026

Purpose

This interpretation clarifies the intended meaning of “building envelope” in the National Building Code – 2023 Alberta Edition (NBC(AE)) as it relates to the expected performance of attic or roof spaces.

Discussion

There are different interpretations of the term “building envelope” being used throughout Alberta, and this poses design and enforcement difficulties, particularly as it relates to attic or roof spaces (see defined term in Code References section).

The National Energy Code for Buildings 2020 (NECB) contains a definition for the term “building envelope,” which has been applied by some in the context of the NBC(AE). Although Sentence 1.4.1.1.(1) of Division A of the NBC(AE) allows terms to have meanings that have been assigned through other Codes or Regulations under the *Safety Codes Act*, the definition of “building envelope” under the NECB reflects a meaning specifically within the context of thermal transmission and does not include all aspects of building envelope performance.

Every material, component and/or assembly that serves or adjoins an attic or roof space has a pivotal role in contributing to the overall performance of the building envelope. For example:

- The roof assembly and exterior wall assemblies of an attic or roof space may not be thermally insulated; however, these assemblies serve the envelope purpose of preventing the ingress of precipitation (rain, snow, etc.) into the building.
- The NBC(AE) requires an unconditioned attic or roof space to be ventilated to promote the transfer of moist air from this space to the exterior air outside the building. Limiting the risk of moisture accumulation, condensation, or ice dams ultimately serves to protect the health of the building occupants and prevent the damage or deterioration of building elements.
- The ceiling assembly between the conditioned space of a building and the unconditioned attic or roof space above it serves as a thermal envelope and also as an air and vapour barrier. This portion of the building envelope aims to limit the leakage of warm, moist air from the conditioned space into the attic or roof space.

Code References

Division A, Sentence 1.4.1.2.(1).

1.4.1.2. Defined Terms

1) The words and terms in italics in this Code shall have the following meanings:

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Attic or roof space means the space between the roof and the ceiling of the top *storey* or between a dwarf wall and a sloping roof.

Conditioned space means any space within a building, the temperature of which is controlled to limit variation in response to the exterior ambient temperature by the provision, either directly or indirectly, of heating or cooling over substantial portions of the year.

NECB 2020, Division A, Sentence 1.4.1.2.(1)

1.4.1.2. Defined Terms

1) The words and terms in italics in this Code shall have the following meanings:

Building envelope means the collection of components that separate *conditioned space* from unconditioned space, the exterior air or the ground, or that separate *conditioned spaces* intended to be conditioned to temperatures differing by more than 10°C at design conditions.

Application

This interpretation applies to buildings of residential occupancy containing one or more dwelling units.

Interpretation

The term “building envelope,” as it relates to the expected performance of an attic or roof space, shall have the following meaning:

1. Building envelope means the collection of materials, components and/or assemblies that:

- separate an attic or roof space from the exterior air or the ground,
- provide the ventilation of an unconditioned attic or roof space to permit the transfer of moisture from the space to the exterior air, or
- separate conditioned space from an unconditioned attic or roof space.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator

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